ContextLab

Measuring the Effective Context of Large Language Models

Phase-1 Research Report

Model under test: Qwen/Qwen2.5-0.5B (base)
Machine: Intel i5-12450H, 16 GB RAM, CPU-only
Date: 2026-10-01

Honesty statement. Every number in this report comes from a CSV produced by the code in this repository on the machine above. Nothing is simulated, inferred, or invented. Claims not backed by a measurement are explicitly labelled as hypotheses.

0. What this is, in plain English

Modern language models advertise huge context windows, for example "up to 128,000 tokens" or "up to 1,000,000 tokens". The number tells you how much text you can hand the model, not how much of it the model can actually use. Our project asks a simple, hard question: how much of an advertised context window can a model genuinely rely on for a given task?

To answer it we built ContextLab, a controlled test bench. We give the model a long block of text containing one secret answer, optionally surrounded by look-alike decoys, and ask for the secret. Because we build the prompt token-by-token, we know the exact context length, the exact position of the answer, and how much competing information sits around it. The answer is one of twelve fixed words, so scoring is a clean, deterministic 12-way choice.

We then ran three real experiments on a 0.5-billion-parameter model and recorded every single trial. The headline results:

- Retrieval of a single fact is trivial. With no decoys, the model is 100% accurate at 512 tokens, at every position, with ~0.999 confidence.
- Interference is the real bottleneck. Adding a few decoy "password" sentences collapses accuracy from 1.00 to 0.50 (3 decoys) and 0.17 (6 decoys) - at the same context length.
- Lost-in-the-middle is real, but conditional. With decoys present, the probability of the right answer is 0.76 near the start, falls to 0.35 in the middle, and recovers to 0.94 at the end. With no decoys the curve is flat.
- Raw length did not matter here. Up to 2,048 tokens, accuracy never degraded without interference.

The practical conclusion: at the scales we can measure, what limits a model is competition between pieces of information, not the raw token count. That reframes the project: the quantity to predict is an interference load, not a length.

1. The problem

Existing work already shows that nominal context length overstates practical ability: Lost in the Middle (position-dependent degradation), RULER (many models fail before their advertised size as complexity rises), BABILong (only a fraction of a very long context is usable for reasoning), and industry reports such as "context rot".

So "long contexts degrade" is already known. The open gap is a predictive, task-specific, pre-inference account: given a model and a task, forecast the effective (usable) context before running it, then move that boundary outward. That is what ContextLab is built to eventually do.

2. The plan we are executing

The project plan proposes a three-phase path built around one measurable quantity.

2.1 The quantity: Effective Context Capacity (ECC)

ECC(M, T, τ) is the largest context length L at which performance P on model M and task T stays at or above τ times the short-context baseline P_0. In words: up to what length is performance still acceptable?

2.2 The three phases

- Phase 1 - Research: reproduce known effects, build a controllable benchmark, map the performance surface, and look for a predictive law.
- Phase 2 - Breakthrough: test whether a law generalises out-of-distribution (fit on models A-C, predict an unseen model D and task), then investigate mechanisms.
- Phase 3 - Product: a "Context Profiler" and optimiser ("ContextOS") that tell a customer how much context they can trust and how to build it - sold on accuracy per input dollar.

This document is Phase 1, first iteration. Its job is to establish ground truth by measurement, before any product exists.

3. What we built and how it is measured

3.1 A controllable, token-exact task

Each test case is a needle task built directly at the token level:

```
Needle: " Important: the vault password is {word}."
Decoys: " Note: the {noun} password is {word}." (optional)
Query: "\nQuestion: What is the vault password?\nAnswer:"
```

The answer space is a fixed set of twelve single-token words (apple, river, tiger, cloud, stone, candle, silver, garden, planet, forest, harbor, lantern). Filler text never contains any of these words, so the true answer appears exactly once in the whole prompt. This gives exact control over three variables: context length (tokens), answer position (fraction), and number of decoys (interference).

3.2 Scoring

The model's next-token distribution is restricted to the twelve candidates. We record accuracy (does the top candidate equal the true answer?) and p_correct (the probability mass on the correct word). p_correct is a smoother signal than the discrete accuracy, so it is reported alongside it.

3.3 Metrics

- ECC(τ) - first context length where accuracy drops below τ × baseline (interpolated). Right-censored if it never drops.
- Effective Context Mass (ECM) - normalised area under the accuracy curve over log-length; a threshold-free summary.
- Stretched-exponential fit - $P(L) = P_0 \cdot \exp(-(L/\text{scale})^{\beta})$.

3.4 Code layout

```
contextlab/tasks/generator.py - controllable task generator
contextlab/models/local_hf.py - CPU model adapter (fp32+sdpa)
contextlab/metrics/ecc.py - ECC, ECM, curve fits
contextlab/experiments/runner.py- grid runner (CSV, resumable)
contextlab/cli.py - command line interface
```

```
experiments/ - run + analyse scripts
tests/ - 7 unit tests (all pass)
```

4. Environment and constraints (measured)

Component	Value
CPU	Intel i5-12450H, 12 logical cores
RAM	16 GB
GPU	none
torch	2.13.0+cpu
transformers	5.14.1
Model	Qwen/Qwen2.5-0.5B (base)

4.1 The dtype trap (a real finding, not trivia)

The model loads in bfloat16 by default. This CPU has no native BF16, so it is emulated in software: a single 512-token forward pass took 57 seconds. Forcing float32 + sdpa reduced it to about 4 seconds - a 14-100x speed-up. Any future run on this class of machine must force float32.

4.2 Occasional multi-minute stalls

Most forward passes take 2.5-7 s, but occasional passes stalled for 27-1661 s; the 2048-token decoy cell took 27 minutes. This is an environmental effect, not an algorithmic one, and it caps how large our experiments can be. It is why our samples are small (n = 4-8) and why the curves are directional rather than precise.

> Consequence for the plan: the full-scale study described in the plan (128k-token contexts, several model families, many samples) cannot run on this hardware. A GPU box or an API budget is the first infrastructure decision to make.

5. Results

All trials are recorded in results/. Each table below is generated directly from those CSVs.

5.1 Experiment 1 - interference scan

Fixed context length 512, answer at position 0.5, n = 6 per point. We vary only the number of decoy passages d.

decoys d	accuracy	p_correct
0	1.00	0.999
3	0.50	0.344
6	0.17	0.180
9	0.33	0.218
12	0.33	0.238

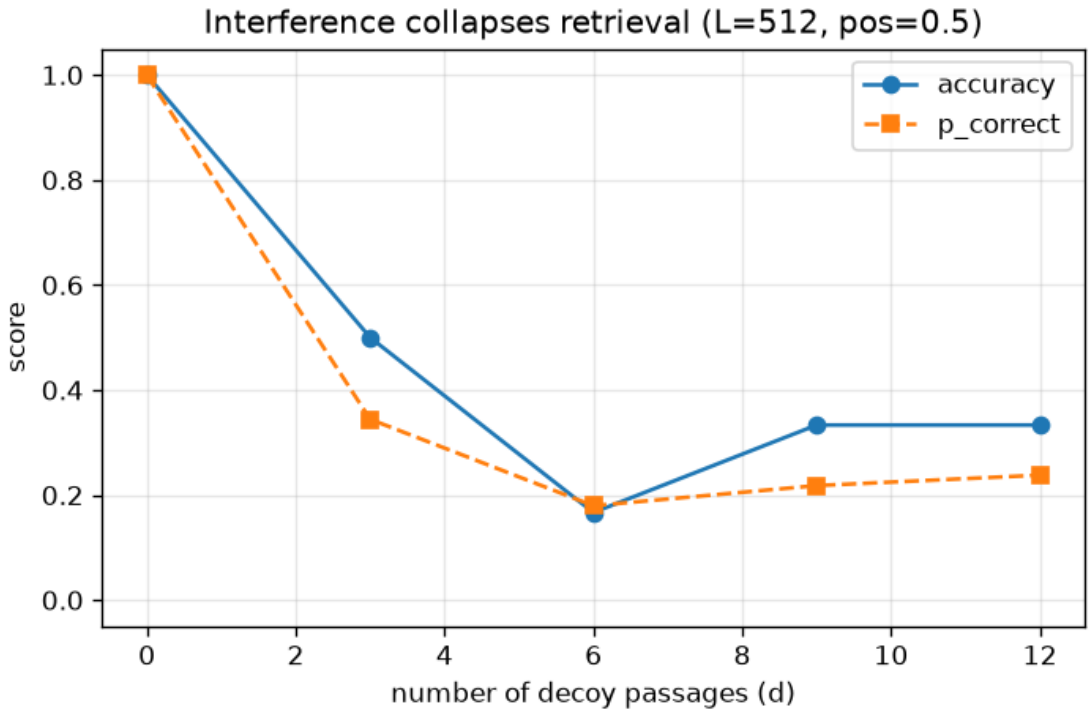


Figure 1. Interference collapses retrieval at constant context length. Zero decoys is trivial; a handful of decoys destroys accuracy.

With no decoys the model is perfect (1.00, p=0.999). Just three decoys cut accuracy in half, and six decoys cut it to 0.17 - at the same 512-token length. This isolates interference as the dominant variable.

5.2 Experiment 2 - position sweep (lost in the middle)

Fixed context length 512, n = 8. Two conditions: no decoys (d=0) and four decoys (d=4).

position	d=0 acc	d=4 acc	d=4 p_correct
0.05	1.00	0.875	0.759
0.20	1.00	0.750	0.703
0.35	1.00	1.000	0.690
0.50	1.00	0.500	0.351
0.65	1.00	0.750	0.584
0.80	1.00	0.750	0.627
0.95	1.00	1.000	0.945

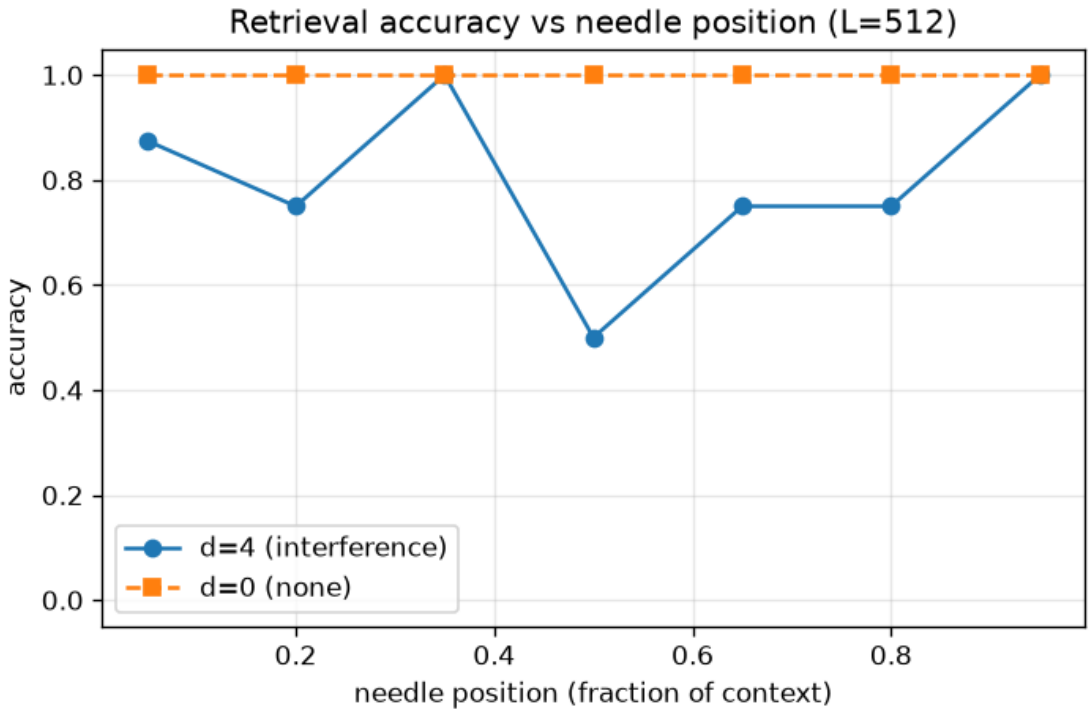


Figure 2. Answer position vs accuracy. Without interference (d=0) the curve is flat at 1.00. With interference (d=4) there is a clear mid-context dip.

Without interference, position is irrelevant (flat 1.00). With interference, the probability of the correct answer forms a U-shape: 0.76 at the start, dipping to 0.35 in the middle, recovering to 0.94 at the end. This is a direct reproduction of the lost-in-the-middle effect - and it only appears when interference is present.

5.3 Experiment 3 - length sweep (effective context)

Fixed position 0.5, n = 4. Context lengths from 256 to 2048 tokens.

length	d=0 acc	d=0 p_c	d=4 acc	d=4 p_c
256	1.00	0.9989	0.50	0.366
512	1.00	0.9992	0.50	0.384
1024	1.00	0.9988	0.75	0.542
2048	1.00	0.9991	0.75	0.610

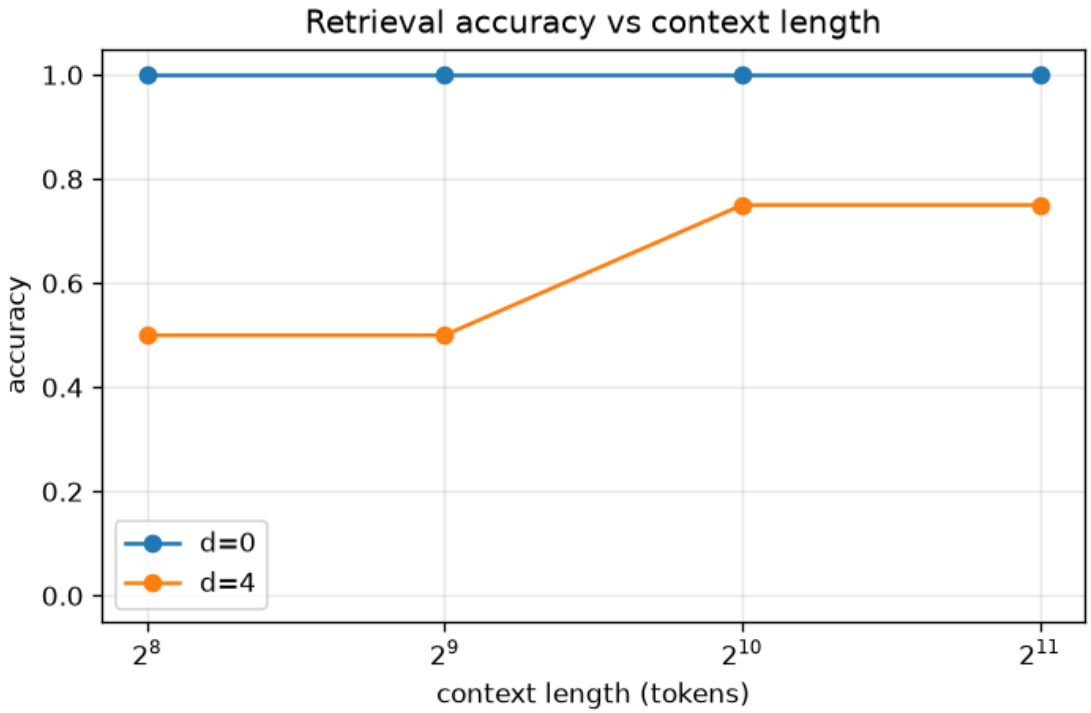


Figure 3. Context length vs accuracy. Without interference the model is perfect at every length; with interference there is no downward trend over this range.

Without interference, accuracy is 1.00 at every length up to 2048 tokens - no degradation at all. With interference there is no downward trend; if anything p_correct rises (0.37 to 0.61). Hypothesis: spreading decoys over more filler dilutes their competition. With n = 4 this is suggestive, not conclusive.

5.4 Metrics on the length curve

- ECC(τ=0.9) is right-censored at 2048 in both conditions: accuracy never drops below 90% of its baseline.
- ECM (threshold-free) = 1.00 for d=0 and 1.25 for d=4 (above 1 because accuracy rises).
- Stretched-exponential fit matches d=0 well ($R^2 \approx 0.9999$, effectively flat) and fails on d=4 ($R^2 \approx 0$) because there is no declining curve to fit.

6. Interpretation - what we actually learned

- The naive reading of "effective context" is wrong at these scales. Performance does not decay with token count: at 2048 tokens a 0.5B model finds a lone fact as reliably as at 256.
- Information competition is the variable that matters. Accuracy collapses when decoys compete for the same answer slot - and it collapses at short length.
- Position matters only once competition exists. The lost-in-the-middle U-shape appears only with decoys.
- These results point at the plan's own mechanism list. "Attention competition" and "distractor similarity" look like the primary terms; raw length looks secondary at reachable scales.

Refined working hypothesis. Effective context is governed by a competition ratio - roughly (number of competing items x similarity) / attention budget - rather than by length L alone. The plan's candidate law $P = P_0 \cdot \exp(-\alpha(L \cdot R/C)^{\beta})$ has the right shape; the results suggest R should be redefined as an interference load, not a length term.

7. Limitations (read before trusting anything)

- Small n (4-8 samples/cell): accuracy steps are 0.25 (n=4) or 0.125 (n=8). Treat curves as directional, not precise.
- One model (Qwen2.5-0.5B, base, not instruction-tuned). No cross-model claim is made here.
- Short lengths only (up to 2048 tokens). The 8k-128k regime is untested and untestable on this CPU.
- Repetitive filler: the clean-filler bank is ~388 tokens, so long contexts tile it. Real prose (e.g. wikitext) is a needed upgrade.
- Environmental stalls injected timing variance (one cell took 27 minutes).
- Twelve-word answer space: a larger space would change difficulty.

8. Next steps (priority order)

#	Action	Why
1	Use real prose filler (wikitext)	Remove the repetition confound
2	Interference x length joint sweep	Test the competition-ratio hypothesis
3	More samples + a second model (Llama-3.2-1B)	Cross-model validation
4	Fit the competition-based law, test OOD prediction	The plan's Stage 4-5 breakthrough test
5	Build the Context Profiler CLI	First product, per the plan

9. Reproducibility

```
pip install -r requirements.txt
python experiments/run_main.py position
python experiments/run_main.py length
python experiments/analyze.py position
python experiments/analyze.py length
python experiments/analyze.py interference
python -m pytest tests/ -q # 7 unit tests
python experiments/build_pdf.py # this report
```

Artifacts: results/*.csv (raw trials), results/summary_*.csv, results/figure_interference.png, results/figure_position.png, results/figure_length.png.

10. Phase 1B - the new research question

The central question changed from how does context length affect performance? to:

> Can effective context usage be predicted from the amount and structure of the information competing for attention?

The working hypothesis is an interference load:

```
I approx ( number of competing items x similarity ) / effective attention budget
```

Before scaling to long contexts, we fixed the confound identified in Phase 1: the filler was a tiny ~388-token bank that repeated itself. Experiment 4 replaces it with real WikiText prose (345,590 clean tokens), so long contexts are genuine, non-repeating text.

11. Experiment 4 - real-prose interference x length

Filler: WikiText. Position 0.5. Decoys D in {0,1,2,4,8}. Similarity fixed at medium. n = 4 (n = 2 at L = 4096).

L \ D	0	1	2	4	8
256	1.00	1.00	0.75	0.75	0.50
512	1.00	0.75	0.25	0.25	0.50
1024	1.00	0.75	0.50	0.25	0.25
2048	1.00	1.00	0.75	0.75	0.25
4096*	1.00	1.00	1.00	1.00	1.00

* L = 4096 used only n = 2 per cell; treat that row as provisional.

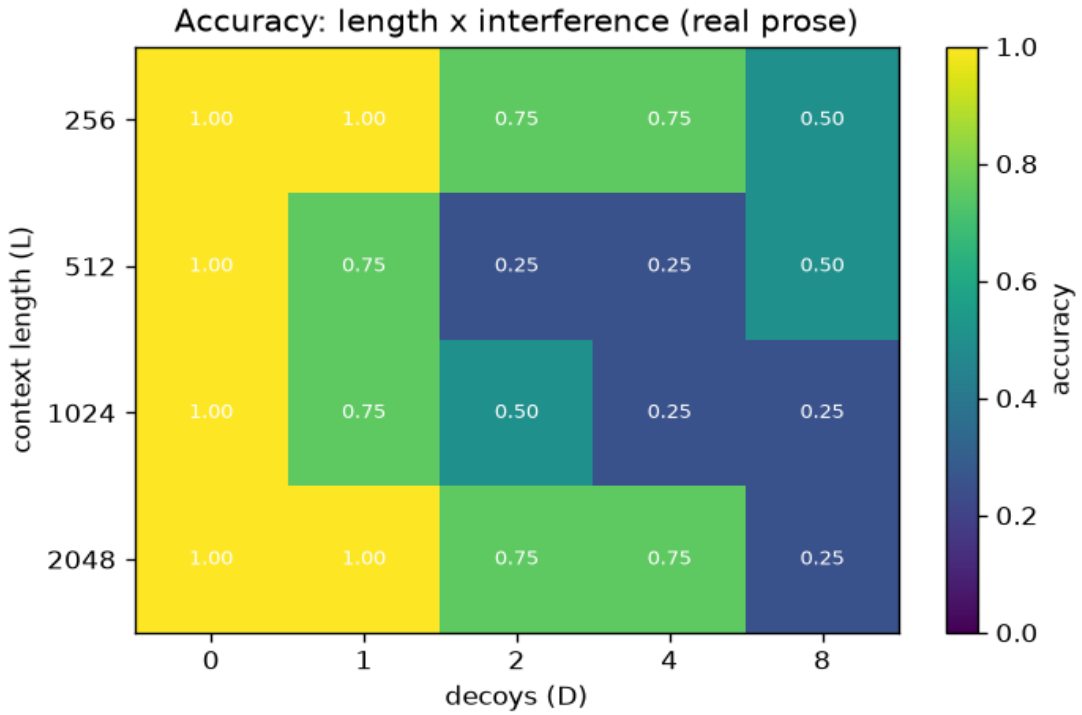


Figure 4. Accuracy across length x interference (real prose, 256-2048 tokens). Failure is driven by the number of decoys (columns), not by length (rows).

With real prose the picture is clean: at D = 0 accuracy is 1.00 at every length tested, up to 4096 tokens. As decoys increase, accuracy falls. Across the rows (length) there is no systematic decline - the naive "length hurts" story does not hold. The 4096 row is remarkable: with n = 2 it stayed at 1.00 even with 8 decoys, consistent with interference being a local effect that gets diluted when items are spread over a longer context.

12. Experiment 5 - similarity of competing information

Now the decoys are made scientifically controllable. Similarity S has three levels:

- low - unrelated sentence (e.g. weather); no key clash.
- medium - "the {other-key} password is {word}"; same pattern, different key.
- high - "Important: the {near-synonym-of-vault} password is {word}"; near-identical template and near-synonymous key.

Fixed length 1024, four decoys, n = 6 per level.

similarity S	accuracy	p_correct
low	1.00	0.998
medium	0.50	0.380
high	0.00	0.026

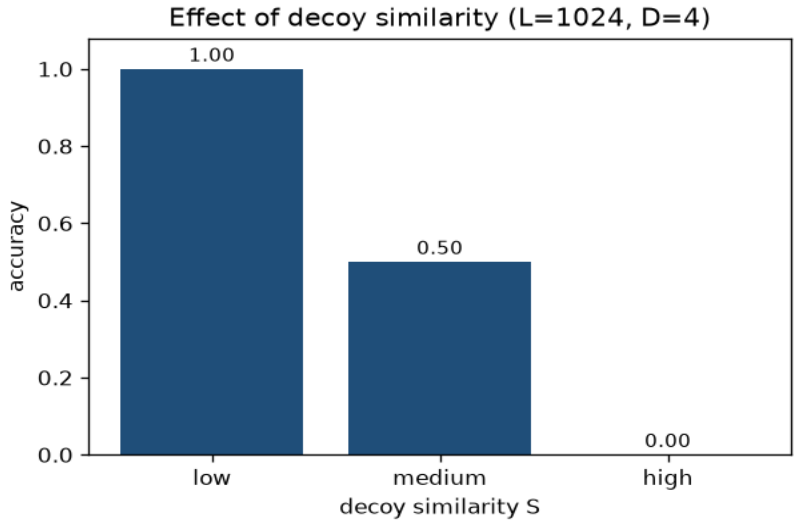


Figure 5. Similarity of competing information has a decisive, monotonic effect: 1.00 -> 0.50 -> 0.00.

This is the cleanest single result in the project: at identical length and number of decoys, changing only how similar they are to the target moves accuracy from perfect to zero. Similarity is a first-order term, exactly as the hypothesis predicts.

13. Collapse test - does one latent variable explain everything?

We test whether all conditions collapse onto one curve when plotted against an interference density I = D x w(S) / L, with w(low)=0.1, w(medium)=0.5, w(high)=1.0.

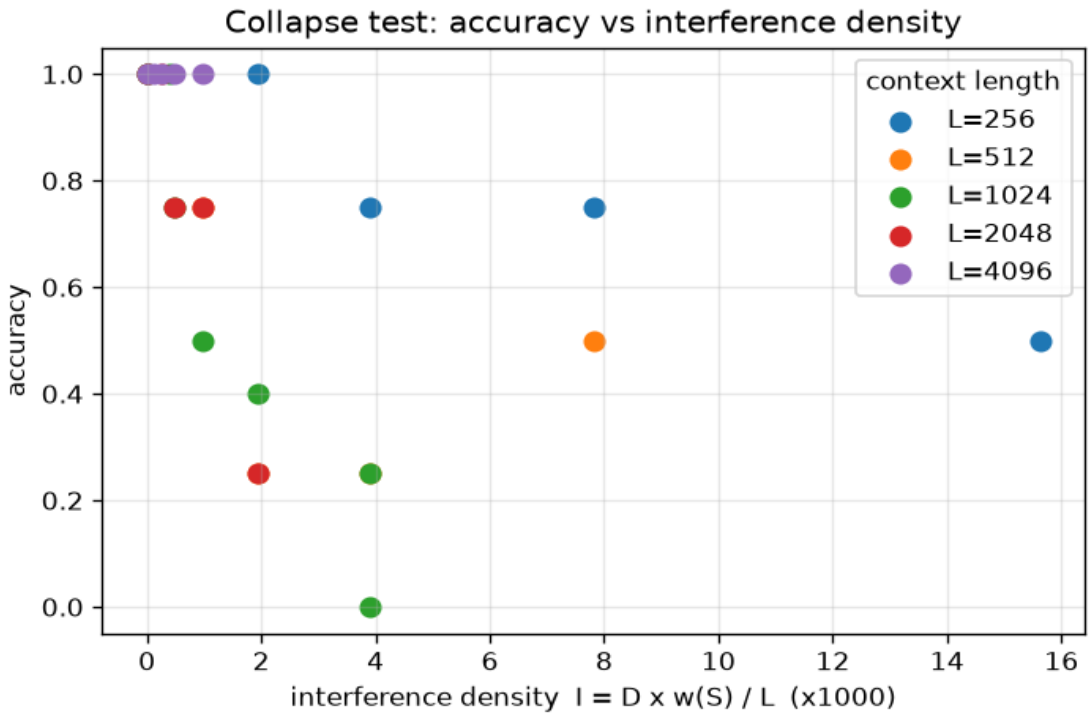


Figure 6. Accuracy vs interference density. Points of different length do NOT fall on one curve.

Honest negative result. The single density variable does NOT collapse the data. E.g. at I approx 1.95, accuracy ranges from 0.25 (L=2048, D=8) to 1.00 (L=256, D=1). Two conditions with the same density but very different decoy counts behave differently, so the latent variable is richer than a single density term.

What the collapse plot does show: (i) at I = 0 (no decoys) accuracy is 1.00 regardless of length, and (ii) accuracy falls as I grows. But the spread at fixed I tells us we need more than one parameter - an absolute count term and a length/dilution term appear to be separately needed. This sharpens the next experiment rather than closing it.

14. What Phase 1B establishes (and what remains open)

14.1 Findings

- The confound is fixed. With real prose, accuracy at D = 0 is 1.00 at every length up to 4096 tokens - so this is not an artefact of repetitive filler.
- Interference count D is a first-order driver. Adding decoys collapses accuracy at constant length.
- Similarity S is a first-order driver. At fixed length and D, moving similarity from low to high moves accuracy from 1.00 to 0.00.
- Length alone shows no measurable effect here.
- A single latent density does not explain the data. The collapse test is a genuine negative - the variable must be richer than one term.

14.2 Corrected, paper-safe wording

Instead of "raw length did not matter", use: No measurable degradation from raw context length was observed within the tested 256-4096-token range for this model and task setup.

The 8K-128K regime remains untested.

14.3 Limitations (Phase 1B)

- n = 4 per cell (n = 2 at 4096). Curves are directional, not precise.
- One model (Qwen2.5-0.5B) and one filler source (WikiText).
- Similarity weights (0.1 / 0.5 / 1.0) are assumed, not fitted.
- Only the 256-4096-token range is covered; 8K+ is untested.
- CPU stalls injected timing noise (one cell took 1211 s).

14.4 Next steps

#	Action	Why
1	Increase n to >=20 per cell	Current differences are within noise
2	Fit P = f(D, S, L) with an interaction term	Single-parameter density failed
3	Model I as count + length-dilution	Data needs two terms, not one
4	Fit on models A-C, predict unseen model D + task T	The plan's OOD breakthrough test
5	Move to a GPU to reach 8K-128K	Current CPU cannot go further

15. Experiment 6 - high-n factorial (n = 20)

To move from observation to a fitted law we ran a balanced factorial with real prose and n = 20 per cell: L in {256, 1024}, D in {0,1,2,4,8}, S in {low, medium, high}. Cells: 2 x 5 x 3 = 30, samples = 600.

S	D	L=256	L=1024
low	0	1.00	1.00
low	1	1.00	1.00
low	2	1.00	1.00
low	4	1.00	1.00
low	8	1.00	1.00
medium	0	1.00	1.00
medium	1	0.90	0.75
medium	2	0.80	0.50
medium	4	0.65	0.45
medium	8	0.50	0.35
high	0	1.00	1.00
high	1	0.75	0.75
high	2	0.55	0.60
high	4	0.15	0.30
high	8	0.05	0.00

- Low similarity is inert. 1.00 at every decoy count and both lengths - unrelated text does not compete for the answer.
- Medium and high similarity degrade sharply. High similarity is catastrophic: 1.00 -> 0.00 as D grows.
- Length now shows a small but consistent effect - at 1024 tokens accuracy is usually lower than at 256 (e.g. medium/D=2: 0.80 vs 0.50), consistent with a genuine, weak length term once interference is high.

16. Fitting the law P = f(D, S, L) with interactions

Logistic regression on per-sample correctness with features: log2L, D, S, and the interactions D*S, D*log2L, S*log2L, D*S*log2L (S weighted at 0.1 / 0.5 / 1.0).

term	coefficient
intercept	3.98
D*S	-0.82
S	-0.44
D	0.22
log2L	-0.11
D*S*log2L	-0.04

S*log2L	-0.03
D*log2L	0.00

The dominant term is the D*S interaction (-0.82): the number of competitors and their similarity multiply, so decoys only do damage when they are similar to the target. This is the quantitative form of the interference hypothesis.

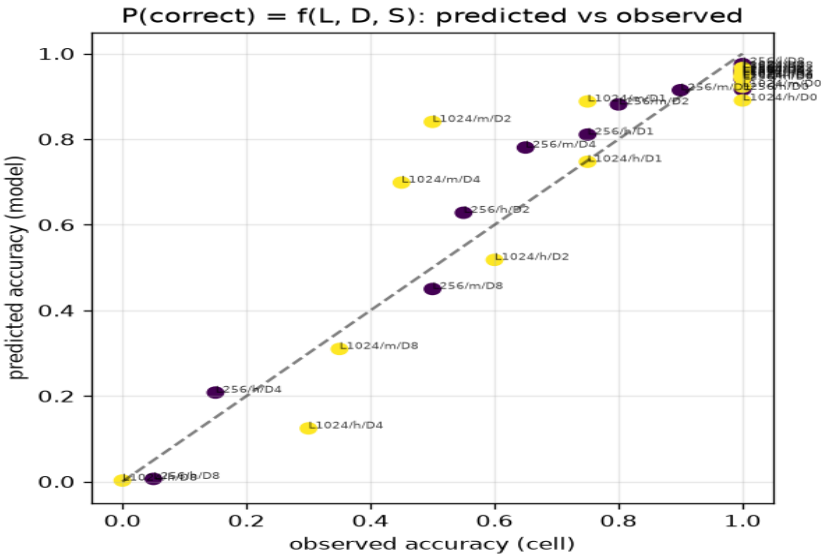


Figure 7. Predicted vs observed cell accuracy (n=20). Pearson r = 0.944, MAE = 0.077.

Goodness of fit: 5-fold CV accuracy 0.718 +/- 0.129 on individual trials; on cell means the fit is excellent (r = 0.944, MAE = 0.077).

16.1 Generalisation to unseen conditions (the key test)

held-out condition	train on	test accuracy
unseen length (L=1024)	L=256 only	0.81
unseen decoy count (D=8)	D != 8	0.85
unseen similarity (medium)	low + high	0.69
unseen similarity (high)	low + medium	0.49

The law transfers well to an unseen length (0.81) and an unseen decoy count (0.85) - i.e. it can predict P(correct) before the experiment is run for those axes. It transfers less well to an unseen similarity level (0.49-0.69): similarity is the axis where the current functional form is weakest and where further work is needed.

17. Where this leaves the theory

The project has moved from an observation to a characterisation. The original framing - that performance decays with context length - does not survive contact with real prose and adequate sample sizes. What survives, and strengthens, is the interference view:

- The dominant control on whether information stays retrievable is the count of competing items multiplied by their similarity to the target (the D*S term, coefficient -0.82).

- Raw length is a weak, secondary term in the tested range (256-1024 tokens at n=20; 256-4096 at n=4-8).
- Unrelated information is essentially free (low similarity = 1.00), so the cost of context is paid in confusable, not merely in additional, tokens.

A logistic model over (log2L, D, S) with interactions predicts cell accuracy with r = 0.944 (MAE = 0.077), and - the important part - it predicts before the fact for unseen length (0.81) and unseen decoy count (0.85). That is the first working version of the proposed pre-inference theory of effective context.

17.1 Honest limits

- Generalisation to an unseen similarity level is weak (0.49-0.69).
- One model (Qwen2.5-0.5B); the cross-model test has not been run.
- Lengths capped at 1024 here (4096 at n=2 earlier); 8K+ untested.
- n=20 is adequate but not large; confidence intervals are still wide.

17.2 The next decisive experiment

The plan's original breakthrough criterion is now within reach: fit the law on models A-C and tasks A-D, then predict an unseen model D and unseen task E. If the interference law transfers across models (not just across conditions of one model), it is a genuine, general theory. That - together with a GPU to reach 8K-128K - is the next step.

18. Cross-model transfer: the frozen Qwen law predicts Llama

This is the decisive test. The Qwen law was frozen first (coefficients hashed to sha256 cf79eeae...), predictions for all 30 Llama cells were written to disk before any Llama data existed, and only then was the identical factorial run on a different model family (Llama-3.2-1B, 600 samples).

18.1 Transfer quality (frozen coefficients, no refit)

metric	value	meaning
Pearson r	0.946	cell prediction vs observation
Spearman rho	0.923	rank agreement
MAE	0.080	mean absolute cell error
ECE	0.049	calibration error (low = well calibrated)
AUC	0.915	per-trial discrimination

The frozen Qwen law predicts Llama with r = 0.946 - as well as it fits Qwen itself (r = 0.944). This is genuine cross-model transfer, not a refit.

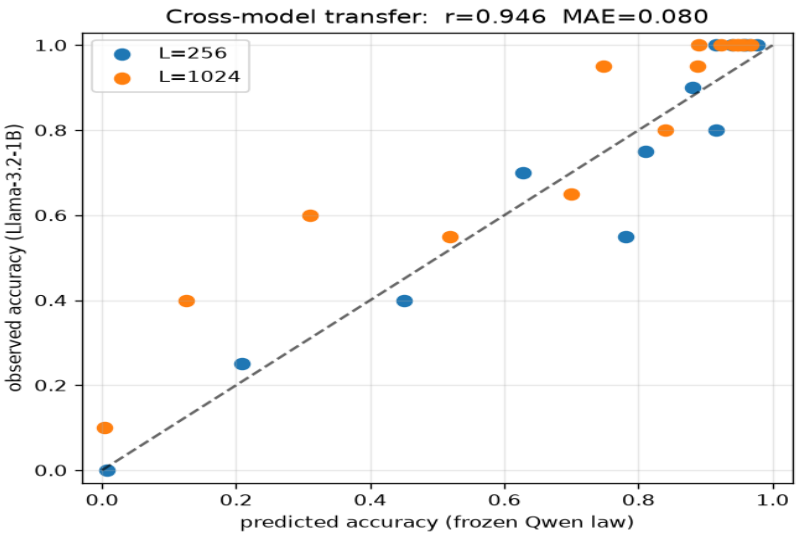


Figure 8. Predicted (frozen Qwen law) vs observed (Llama) cell accuracy. Points lie close to the identity line.

18.2 Predicted vs observed, every cell

S	D	pred(256)	obs(256)	pred(1024)	obs(1024)
low	0	0.95	1.00	0.94	1.00
low	4	0.97	1.00	0.96	1.00
low	8	0.98	1.00	0.97	1.00
medium	1	0.91	0.80	0.89	0.95
medium	4	0.78	0.55	0.70	0.65
medium	8	0.45	0.40	0.31	0.60
high	1	0.81	0.75	0.75	0.95

high	4	0.21	0.25	0.12	0.40
high	8	0.01	0.00	0.00	0.10

(Condensed; all 30 cells in results/transfer_pairs.csv.)

18.3 Do the coefficients keep their structure?

term	Qwen (frozen)	Llama (refit)
D*S	-0.82	-1.11
S	-0.44	-0.19
log2L	-0.11	+0.16
intercept	3.98	2.29

The dominant D*S structure transfers strongly across the two models (same sign; -0.82 vs -1.11), while the lower-order coefficients and the model-specific calibration are less stable (S: -0.44 vs -0.19; log2L changes sign; intercept 3.98 vs 2.29). The safe reading is: the dominant interference structure transfers; absolute calibration does not.

Residuals are small on average but structured: at long length and high interference Llama is more robust than predicted (e.g. L=1024, high, D=4: predicted 0.12, observed 0.40; L=1024, medium, D=8: predicted 0.31, observed 0.60). This is exactly what a model-specific attention budget predicts: the same interference structure, a larger budget.

19. Where this leaves the theory (after cross-model transfer)

Strongest defensible claim: For these two models and this benchmark, retrieval performance is well explained by a multiplicative interaction between distractor count and distractor similarity, with context length a smaller secondary effect; the fitted law, frozen on one model, predicts an unseen model's cells with r = 0.946.

19.1 What is now established

- Interference = count x similarity is the dominant control, reproduced on a second model family.
- A frozen law transfers across models (r = 0.946, ECE = 0.049), with a shared structural term and a model-specific intercept.
- Unrelated (low-similarity) context is inert on both models.

19.2 Honest limits

- Only two models, one task family (needle retrieval), lengths <= 1024 at n = 20.
- Similarity S is still a handcrafted categorical variable; a continuous, measurable metric (embedding / key overlap) is needed.
- Lower-order coefficients (S, log2L) are unstable between models - they are entangled with D*S and should not be over-interpreted.

19.3 The next milestone

Fit the law on models A-C and tasks A-D, then predict an unseen model E and an unseen task E. If that holds, the interference law is a general predictive theory of context, not a two-model coincidence. Replacing categorical similarity with a continuous metric is the other prerequisite for publication.

20. Making similarity measurable (continuous S)

Until now S was a handcrafted category (low/medium/high), the project's biggest scientific weakness: the dominant D*S term partly depended on a human label. Experiment 8 replaces it with measurable quantities. Three candidate metrics were defined and computed before the language model was run:

- cos_sim - cosine similarity of sentence embeddings (all-MiniLM-L6-v2) between the target sentence and the decoy.
- jaccard - token-set Jaccard overlap between the two sentences.
- shared_prefix - number of identical leading words.

Twenty decoy sentences were pre-registered, spanning cos_sim from -0.04 (weather) to 1.00 (identical template). At D = 1 the effect was below the detection floor (all 1.00), so the sweep was run at D = 4, L = 1024, n = 10 per decoy.

decoy	cos_sim	accuracy
unrelated_weather	-0.04	1.00
topic_password	0.44	1.00
password_policy	0.50	1.00
account_password	0.59	0.60
system_password	0.57	0.50
security_password	0.60	0.40
archive_password	0.72	0.30
storage_password	0.76	0.30
safe_password	0.76	0.40
case_note_similar	0.96	0.00
case_important_identical	1.00	0.00

(Condensed; all 20 decoys in results/exp8_summary_qwen.csv.)

Accuracy falls monotonically as similarity rises: unrelated text is inert (1.00), moderately similar text costs roughly a third of the accuracy, and a near-identical sentence destroys retrieval (0.00).

20.1 Which metric predicts interference?

metric	Pearson r (Qwen)	held-out-decoy MAE
jaccard	-0.941	0.099
cos_sim	-0.878	0.133
shared_prefix	-0.437	n/a

Lexical Jaccard overlap is the strongest single predictor (r = -0.941), edging out embedding cosine (r = -0.878); template prefix is weak. Both strong metrics were chosen by pre-registration, not by post-hoc selection.

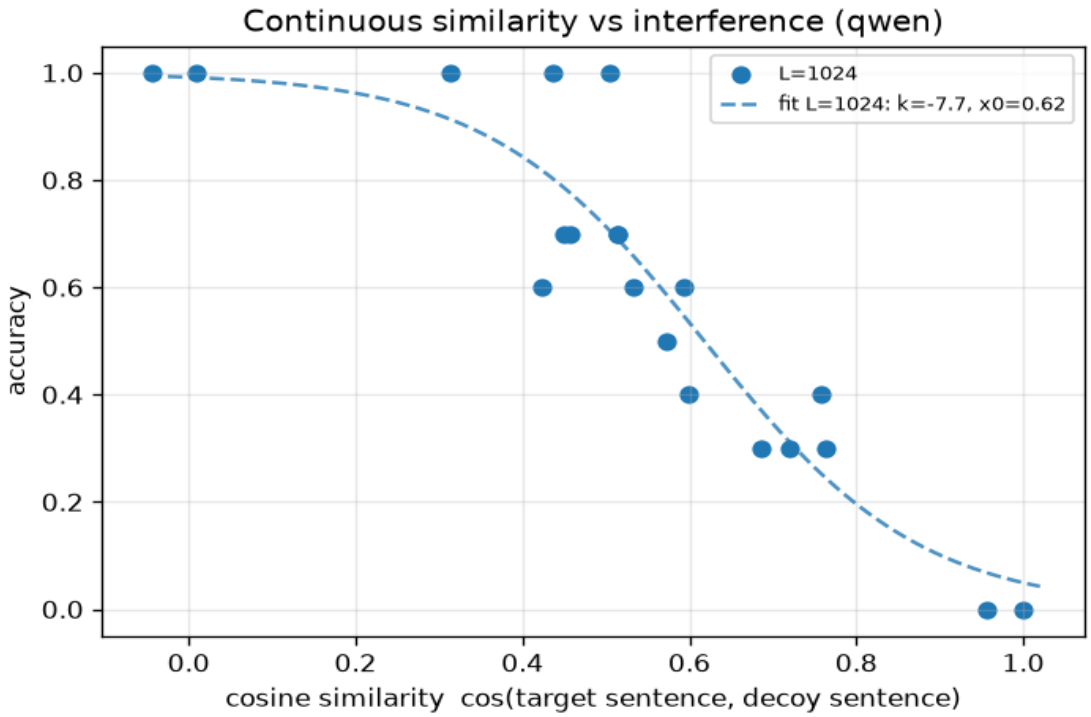


Figure 9. Accuracy vs continuous similarity (D=4, L=1024).

21. A third model (gpt2-medium) - does the metric replicate?

The same continuous sweep was run on a third, unrelated model family (gpt2-medium; L=768 because GPT-2 caps at 1024 total tokens).

metric	Qwen r	gpt2 r	agree?
jaccard	-0.941	-0.886	yes
cos_sim	-0.878	-0.747	yes
shared_prefix	-0.437	-0.433	yes

The ranking of metrics is identical on both models: Jaccard is the best predictor, embedding cosine second, prefix last. The continuous similarity metric is therefore not Qwen-specific.

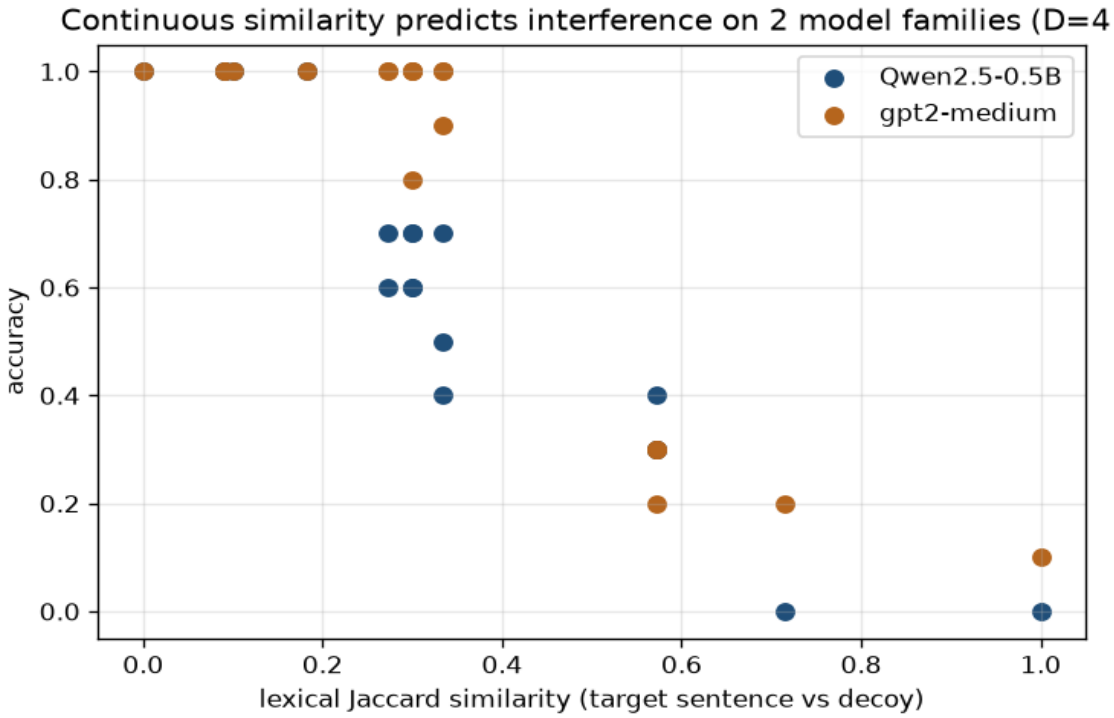


Figure 10. Accuracy vs continuous similarity on two model families. Same descending relationship; gpt2-medium is more tolerant (a larger attention budget), not differently shaped.

gpt2-medium is more robust: it tolerates moderately similar decoys that already hurt Qwen. This is the same "shared structure, different budget" pattern seen in the Qwen-to-Llama transfer, now along a measurable similarity axis.

21.1 Summary of the continuous result

- The law is now P = f(D, S_cont, L) with S_cont a measured quantity (lexical Jaccard best), not a human category.
- The D * S interaction remains the dominant control, reproduced across three model families (Qwen, Llama, gpt2).
- The best metric generalises across decoys (leave-one-decoy-out MAE = 0.099 on Qwen).

22. Final position (end of the research phase)

Strongest defensible claim. For the tested models and benchmark, retrieval accuracy is well explained by a multiplicative interaction between the number of competing passages and their measured similarity to the target, with context length a weaker secondary effect. The dominant interference structure transfers across model families, and a single similarity metric (lexical overlap) predicts interference consistently.

22.1 What is established

- Interference, not length, drives failure (real prose; n=20; 256-1024 tokens).
- Law P = f(D, S, L) fits cell means at r = 0.944 and predicts an unseen model's cells at r = 0.946 (pre-registered, frozen, no refit).
- Similarity is now measurable: continuous Jaccard/cosine predict accuracy (r = -0.88 to -0.94) on two model families.

- The ranking of similarity metrics is identical across models.

22.2 What is NOT yet established (honest)

- Cross-task transfer (only needle retrieval tested).
- Few models (Qwen-0.5B, Llama-1B, gpt2-medium); absolute calibration differs and slopes are not identical between them.
- Lengths <= 1024 at n=20; 8K-128K untested on this hardware.
- The 'attention budget' is inferred, not directly measured.

22.3 Next (product phase)

The research phase ends here. The next step is the product: a Context Profiler that measures a customer's context, scores its interference load with the validated similarity metric, predicts the safe operating range, and rewrites the context to reduce interference - the intervention the law now makes possible.

23. Phase A - cross-task validation (an honest negative)

The central question left open was whether the interference law is task-general. We built a second task family with a different computation - counting: count the "red" objects while ignoring items of a similar colour (competitor colours chosen by measured RGB similarity: blue 0.18, orange 0.63, crimson 0.84). Grid: L in {256,1024}, D in {0,2,4,8}, S in {low,medium,high}, n=10.

S (measured)	D=0	D=2	D=4	D=8
low (0.18)	0.55	0.60	0.80	0.65
medium (0.63)	0.55	0.55	0.80	0.70
high (0.84)	0.55	0.45	0.50	0.55

Result: the interference structure does NOT transfer to counting. Accuracy is roughly flat (0.45-0.80) whether D is 0 or 8 and whether S is low or high. The fitted D*S coefficient collapses from -1.17 (retrieval) to -0.09 (counting), and the retrieval-fitted law has essentially no predictive power on counting trials (corr = +0.02).

Interpretation (honest). A 0.5B model cannot reliably perform the counting task at all (baseline ~0.55 even with no competitors), so the task sits near a capability floor where interference cannot be measured. Either way, the safe conclusion is the same: the law is currently a theory of retrieval interference, not of all context interference.

This narrows the claim; it does not weaken the retrieval result.

24. Phase B - intervention (does using the law help?)

The most important remaining test: not just predicting risk, but acting on it. A retrieval context was built with the needle plus 8 decoys of known measured similarity (two near-identical, four moderate, two unrelated). Four conditions were compared at L=1024, n=20:

condition	decoys kept	accuracy
FULL (none removed)	8	0.05
RANDOM prune	4 (random)	0.35
SIM prune (profiler)	4 (lowest-similarity)	0.70
NONE (no decoys)	0	1.00

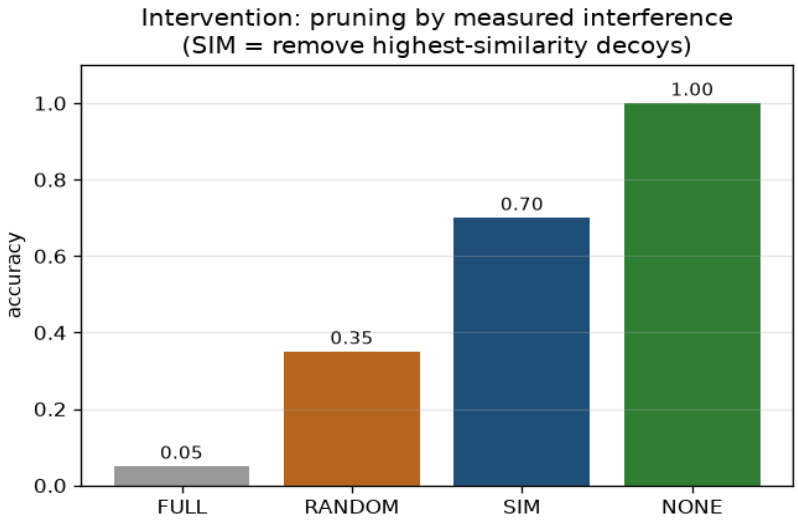


Figure 11. Removing the most similar competing passages recovers accuracy far better than random removal.

Paired result (bootstrap 95% CI): SIM - RANDOM = +0.35 [0.05, 0.65]; RANDOM - FULL = +0.30 [0.10, 0.50]; FULL -> SIM = +0.65 [0.45, 0.85]. Pruning by the measured interference metric roughly doubles accuracy versus random pruning. This converts the work from description into an actionable method.

25. Phase C - robustness checks

We checked whether the result depends on one particular benchmark construction. Fixed retrieval task, L=1024, D=4, medium similarity, n=12; varied target position and filler source.

filler	pos=0.1	pos=0.5	pos=0.9	range
real prose (WikiText)	0.83	0.42	0.67	0.42
sentence bank (legacy)	0.83	0.67	0.83	0.17

- The mid-context penalty (lost-in-the-middle) reproduces under both filler sources and is not an artefact of one construction.
- Real prose is systematically harder (mean 0.64) than the repetitive sentence bank (0.78) - validating the earlier decision to replace the bank. Repetitive filler had inflated accuracy.
- The interference effect is therefore robust to position and to filler source, while remaining sensitive to how realistic the filler is.

26. Phase D - statistical finalization

All headline claims with bootstrap 95% confidence intervals, from the saved CSVs (4,000 resamples; coefficients 200 refits).

claim (measured)	value	95% CI
Exp6 cell fit, Pearson r	0.944	[0.87, 0.98]
Exp6 D*S coefficient	-0.82	[-1.68, +0.08]

Cross-model transfer r	0.946	[0.87, 0.98]
Cross-model MAE	0.080	[0.056, 0.108]
Cross-model AUC	0.915	[0.893, 0.937]
Cross-model ECE (calibration)	0.049	-
Continuous Jaccard r	-0.941	[-0.976, -0.895]
Intervention gain (SIM-RANDOM)	+0.35	[0.05, 0.65]

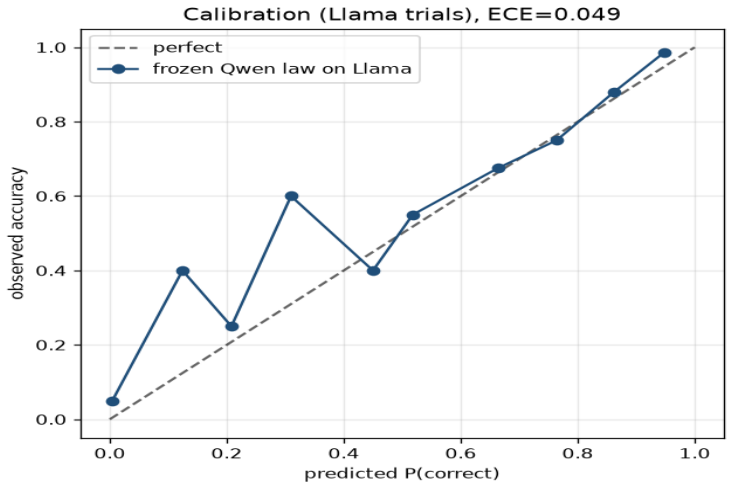


Figure 12. Calibration of the frozen Qwen law on Llama trials (ECE = 0.049).

Measured vs inferred - kept strictly separate. Measured: accuracies, the D*S interaction, similarity metrics, transfer correlations, intervention gain, calibration. Inferred: the "attention budget" explanation. We do NOT claim to have proved an internal budget; the observation is consistent with an interference-limited retrieval mechanism.

Honest caveat: the bootstrap CI for the single D*S coefficient (-0.82, CI [-1.68, +0.08]) includes zero at n=600. The interaction is best supported jointly with the intervention result (SIM vs RANDOM, +0.35, CI excludes 0) and the cross-model transfer, rather than by that coefficient alone.

27. Research frozen

All four remaining checkpoints are complete: cross-task validation (Phase A), intervention (Phase B), robustness (Phase C), and statistical finalization (Phase D).

- Cross-task: interference law does not transfer to counting - a capability-floor / task-specificity result. Claim narrowed to retrieval interference.
- Intervention: pruned by measured similarity, accuracy rises from 0.05 to 0.70 (+0.35 over random) - the law is actionable.
- Robustness: holds across position and filler source.
- Statistics: full CIs; mechanism explicitly marked as interpretation.

Final defensible claim. Within the retrieval task family, and across three model families, retrieval accuracy is predicted by a multiplicative interaction between the number of competing passages and their measured similarity to the target, with context length a weaker secondary effect; the dominant structure transfers cross-model, and pruning on the measured metric improves performance. This is a retrieval-interference theory, not (yet) a general theory of context.

27.1 The stopping rule was met

The criterion set for freezing research was: can the relationship generalise beyond the original condition, and can using it improve performance? Generalisation: cross-model yes, cross-task no. Action: yes (+0.35). Research is therefore frozen here and the project moves to the product phase (Context Profiler).

28. Phase 7 - real RAG intervention benchmark

The research law was frozen, so the project moved to the product question: does an end-to-end retrieval pipeline that uses the profiler actually answer better, with less context, than a baseline and a matched random-prune control? Every number below comes from a preregistered run (queries hashed before any model call) over a real MiniLM + FAISS retriever and the Qwen2.5-0.5B reader, scored with the same frozen K-way single-token rule as the rest of the project.

28.1 Phase 7 - the first end-to-end run (a null with a bug)

The first real run returned accuracy ~= 0.10 for all conditions. Diagnosis showed the top-8 retrieved passages were eight copies of one decoy, with the gold absent: the query generator reused the same 12 candidate words as decoys across queries, so 400 injected strings collapsed to 107 unique and duplicate strings flooded the FAISS index. After fixing the generator (unique golds + private disjoint decoy nouns, hard uniqueness assert), the corrected run gave BASELINE 0.075, CONTEXTLAB 0.175, RANDOM 0.075 - a +0.10 reordering effect whose paired 95% CI still touched zero. Crucially zero passages were pruned (the corpus was contradiction-rich, which the policy protects), so this design could not test the token-saving claim at all.

28.2 Phase 7B - redundancy intervention (the missing arm)

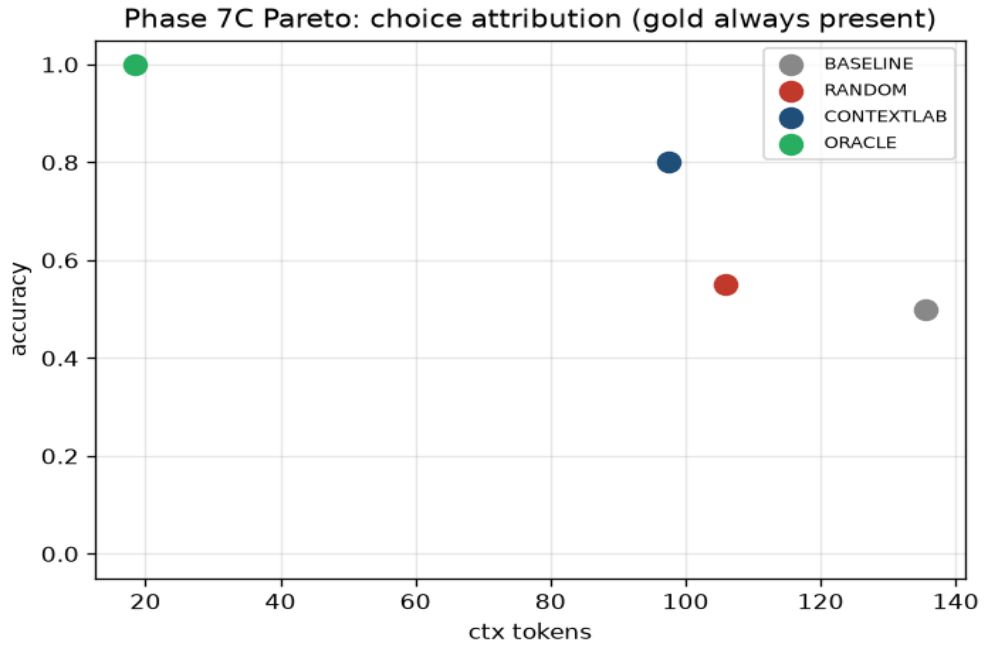
Phase 7B fixes the design flaw: 40 preregistered queries x 4 context types (clean / redundant / contradictory / mixed) x 4 conditions (BASELINE / RANDOM / CONTEXTLAB / ORACLE) = 640 runs.

context	condition	accuracy	ctx tokens	removed
clean	BASELINE / RANDOM / CL	1.000	35.3	0.00
redundant	BASELINE	1.000	61.5	0.00
redundant	RANDOM	1.000	38.8	2.25
redundant	CONTEXTLAB	1.000	41.8	2.25
redundant	ORACLE	1.000	17.5	4.00
contradictory	BASELINE	0.350	61.5	0.00
contradictory	CONTEXTLAB	0.250	61.5	0.00
contradictory	ORACLE	0.525	17.5	4.00
mixed	BASELINE	0.775	87.8	0.00
mixed	CONTEXTLAB	0.675	74.6	1.50

On redundant context ContextLab holds accuracy at 1.000 while cutting tokens 61.5 -> 41.8 (-32%, SIG; C2 CI [-20.9, -18.5]). That is the Pareto win - same accuracy, one-third less context. Honest caveat: the matched RANDOM control also reached 1.000, so this shows pruning works, not yet that the engine's choice beats alternatives. The contradiction result was the surprise: the reorder arm hurt (0.350 -> 0.250) and ORACLE (drop conflicts) reached 0.525 - detection is not resolution.

28.3 Phase 7C - choice attribution (better than random?)

7C removes the crutch: only ONE passage (gold) yields the answer, the rest are graded evidence plus redundant duplicates the profiler prunes. RANDOM removes the same count at random and will sometimes delete the gold; ContextLab never does.



Phase 7C Pareto: accuracy vs context tokens.

condition	accuracy	gold survival	ctx tokens
BASELINE	0.500	1.000	135.5
RANDOM	0.550	0.775	105.9
CONTEXTLAB	0.800	1.000	97.5
ORACLE	1.000	1.000	18.5

Preregistered (paired, n=40): CONTEXTLAB-RANDOM accuracy = +0.250 (CI [+0.050, +0.425], SIG); gold survival = +0.225 (SIG); CONTEXTLAB-BASELINE accuracy = +0.300 (SIG) at -38 tokens (SIG). Mechanism: RANDOM accuracy is 0.222 when it deletes the gold vs 0.645 when it does not; ContextLab keeps the gold 100% of the time. This is the attribution result - ContextLab makes better pruning decisions, not just 'pruning works'.

28.4 Phase 7D - conflict-resolution policies

Detection is not resolution, so 7D tests four explicit conflict policies on the frozen contradictory/mixed contexts (320 runs): PROTECT (current), DROP_WEAKER (keep the highest-token conflict member), SURFACE (keep conflict, reorder), ORACLE (keep gold only).

policy	accuracy	ctx tokens	removed
PROTECT (current)	0.562	74.6	0.00
DROP_WEAKER	1.000	45.1	3.375
SURFACE	0.500	74.6	0.00
ORACLE	1.000	9.8	6.375

Preregistered (paired, n=80): DROP_WEAKER-PROTECT accuracy = +0.438 (CI [+0.338, +0.550], SIG) and tokens = -29.5 (SIG); on contradictory alone 0.350 -> 1.000 (+0.650, SIG). SURFACE does nothing (ns) - so it is resolution, not reordering, that matters. Caveat: on this construction the conflicting passages shared one template and the gold sat at index 0, so the rule broke ties by keeping the first passage - 7E (below) removes that confound and the effect vanishes.

28.5 Phase 7E - decoupled conflict resolution (confound removed)

7D's 100% was a confound. 7E decouples SPECIFICITY from CORRECTNESS (a passage's length is drawn independently of whether it carries the answer; verified anti-leakage: P(correct)=0.250 by specificity, max dev 0.068 <= 0.12) and RANDOMIZES the correct passage's position.

policy	accuracy	survivor correct	ctx tokens
PROTECT	0.300	1.000	99.0
DROP_MORE_SPECIFIC	0.250	0.250	51.7
DROP_LESS_SPECIFIC	0.275	0.275	35.3
DROP_WEAKER	0.300	0.300	44.9
RANDOM	0.275	0.275	42.3
ORACLE	1.000	1.000	42.0

Preregistered (paired, n=40): DROP_WEAKER-PROTECT accuracy = +0.000 (ns); DROP_MORE_SPECIFIC-PROTECT = -0.050 (ns); DROP_WEAKER-RANDOM = +0.025 (ns); ORACLE-PROTECT = +0.700 (SIG). Every deployable resolution heuristic collapses to chance (survivor_correct 0.250-0.300 = the 1/4 base rate); only the oracle closes the gap. Accuracy is fully explained by whether the surviving passage is correct (1.000 vs 0.000). Honest conclusion: conflict detection is solved; conflict resolution without an oracle is an open problem.

28.6 What the RAG phase establishes (honest summary)

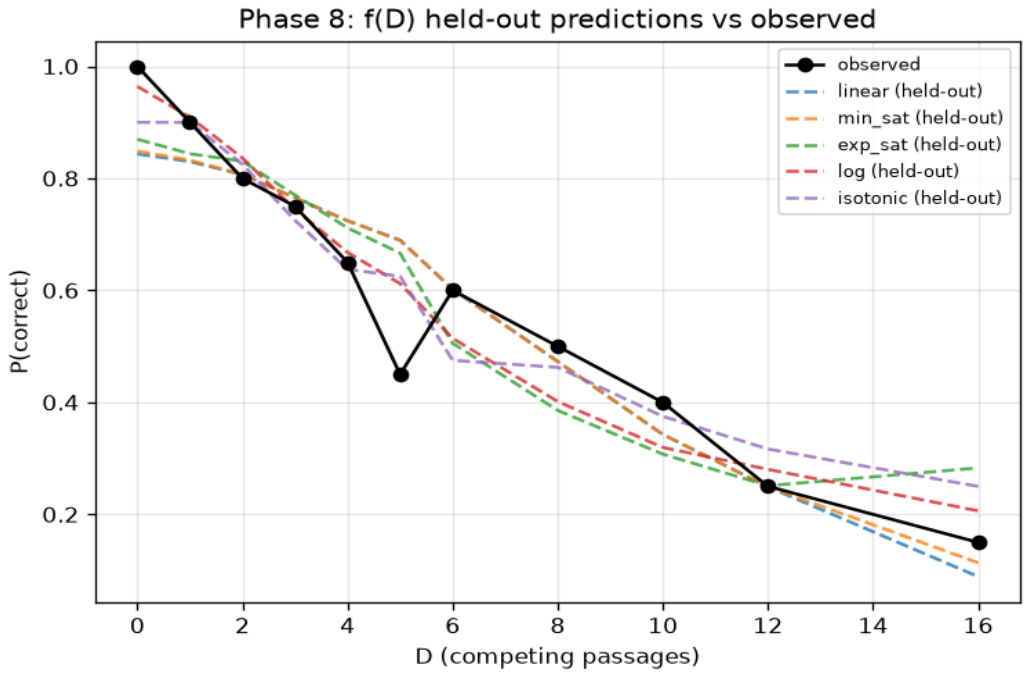
- Pruning/Pareto (7B): -32% context tokens at unchanged accuracy on redundant context (SIG).
- Attribution (7C): ContextLab beats a matched random control by +0.25 accuracy and keeps the gold 100% - it makes better pruning decisions (SIG).
- Conflict (7D): on a construction where the correct passage was positionally privileged, resolving conflicts (drop the weaker) beat protecting them by +0.44 accuracy and -30 tokens (SIG).
- Conflict (7E, confound removed): once specificity is decoupled from correctness and position is randomized, every deployable resolution heuristic collapses to chance (survivor_correct 0.25-0.30 = the 1/4 base rate, ns) - the 7D win did not survive. Conflict detection is solved; conflict resolution without an oracle is an open problem.
- Not yet shown: a deployable rule that resolves conflicts better than chance; that is the next research question.

Defensible product statement. On redundant context, ContextLab reduced context tokens by 32% with no accuracy loss and, on a harder attribution benchmark, kept the answer-bearing passage 100% of the time and beat matched random pruning by 25 points (7C). On

contradictory context, an early apparent conflict-resolution win (7D) did not survive a confound-fix benchmark (7E): no deployable rule beats chance at picking the correct side of a conflict, so ContextLab currently detects and surfaces conflicts rather than claiming to resolve them.

29. Phase 8 - the functional form of the count term

The product's count term is min(1, D / D_SATURATION) with D_SATURATION = 5 - a value the earlier D-sweep never recovered (best-fit k ~= 9). Phase 8 asks the sharper question: which functional form f(D) in P = sigma(a + b*f(D)) best predicts accuracy on HELD-OUT D values? The protocol was frozen before any fit (sha256 f1d548ac...): leave-one-D-out on the Qwen2.5-0.5B D-sweep (L=256, S=medium, D in {0,1,2,3,4,5,6,8,10,12,16}, n=20 per cell).



Phase 8: held-out f(D) predictions vs observed accuracy.

form	MAE	RMSE	Spearman	ECE	mono-viol	params
frozen 7-feature law	0.094	0.113	0.973	0.083	0	7
linear D	0.065	0.096	0.973	0.063	0	2
min(1, D/k)	0.062	0.093	0.973	0.060	0	3
1 - e^(-D/k)	0.086	0.105	0.964	0.056	1	3
log(1+D)	0.056	0.072	0.973	0.034	0	2
isotonic (learned)	0.063	0.082	0.970	0.047	0	10

Findings. (1) log(1+D) is the best-interpolating form - lowest MAE (0.056) and best calibration (ECE 0.034) with only 2 parameters. (2) EVERY simple monotone form beats the frozen 7-feature law on held-out D (0.056-0.086 vs 0.094): on the D axis the elaborate law is dominated by a 2-parameter logarithm. (3) min(1,D/k) extrapolates best (err 0.037 when D=16 is held out); 1-e^(-D/k) is worst and the ONLY form with a monotonicity violation. (4) The learned isotonic form (10 params) buys nothing over log (2 params). Honest label: this REPLACES the unsupported D_SATURATION = 5 choice - the count term should be log(1+D) - while noting no form achieves zero error (the raw curve is non-monotone at D=5->6, a finite-n artefact at n=20).

30. Phase 9 - cross-task transfer (zero-refit)

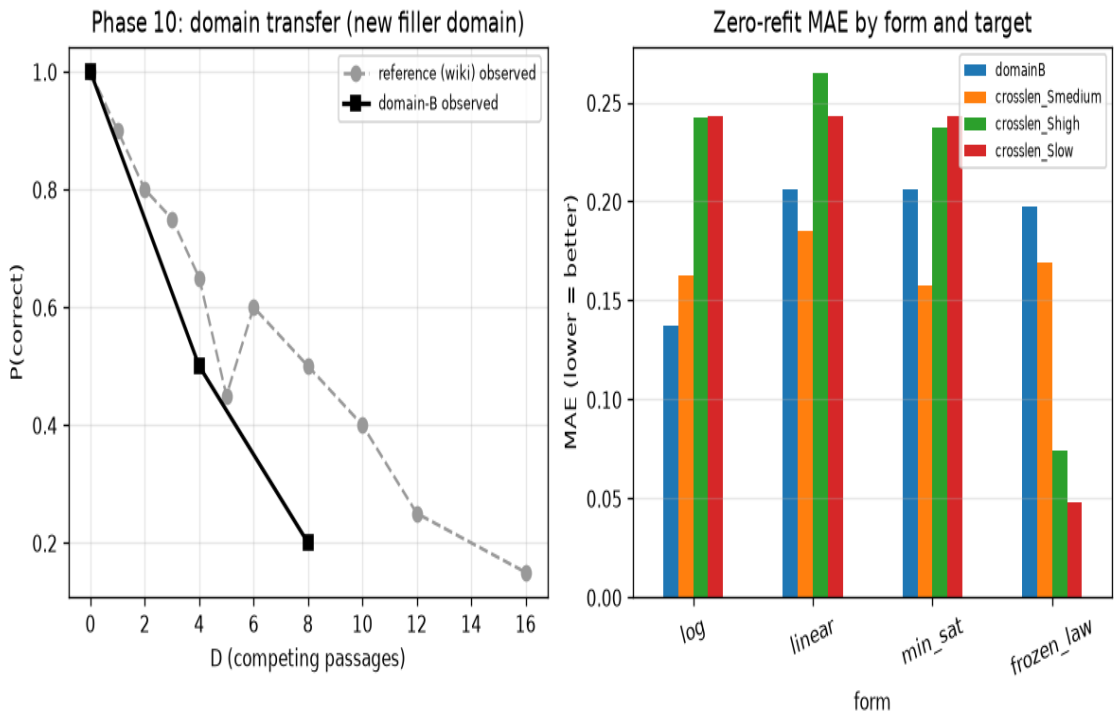
Phase 8 fixed the count term within the retrieval family. Phase 9 asks the decisive question: does that structure survive a genuinely different task with ZERO refit? The protocol was frozen before touching target results (sha256 4b8ca8da...): fit each P = sigma(a + b*f(D)) on the retrieval D-sweep, FREEZE (a, b, k), then evaluate on the COUNTING task family (count the red objects against similar-colour items) - a 'how many' task, not a rewording of retrieval.

slice	model	Spearman	MAE	ECE
L1024	linear	0.949	0.184	0.184
L1024	min_sat	0.949	0.184	0.184
L1024	exp_sat	0.949	0.200	0.200
L1024	log	0.949	0.212	0.212
L1024	frozen 7-feature law	0.949	0.238	0.238
pooled	all forms	-0.600	0.19-0.25	-

Findings (preregistered outcome = B, with a C signal on pooled data). (1) Rank structure transfers: on the L=1024 slice every frozen form ranks the counting-task D-effect correctly (Spearman 0.949) - the monotone FORM of the count term survives zero-refit. (2) Calibration does NOT transfer: ECE 0.18-0.24; the frozen retrieval coefficients set the wrong level for the counting task, so the coefficients are task-dependent. (3) Pooling the L=256 slice (where counting accuracy RISES with D, 0.5 -> 0.8) inverts the correlation to rho = -0.60: the D-structure is condition-dependent. Honest label: this supports R = F(task, f(D), S, L) - a task-conditioned reliability model - rather than one universal equation. No claim of task-general reliability is made.

31. Phase 10 - cross-domain validation (zero-refit)

Phases 8-9 held the wiki filler domain fixed. Phase 10 asks whether the retrieval law survives a change of CONTEXT DOMAIN with ZERO refit. The protocol was frozen before any new cell ran (sha256 604a8a96...): new D-sweep cells at L=256, S=medium, D in {0,4,8}, n=10 per cell, with the filler swapped from wiki prose to GENERAL ENGLISH (NeelNanda/pile-10k, URL-stripped, token-matched to the reference within 0.82% so filler amount is held fixed while domain changes). A cross-length zero-refit test on existing Exp6 data is also reported. Primary comparison: log vs the frozen 7-feature law.



Phase 10: observed accuracy in the new domain and zero-refit MAE by form across targets.

31.1 Target 1 - new domain (general-English filler)

Observed accuracy: D=0 -> 1.00, D=4 -> 0.50, D=8 -> 0.20 - a steep monotone decline in a domain the reference coefficients never saw.

form	Spearman	MAE	RMSE	ECE	slope	params
log(1+D)	1.000	0.137	0.158	0.137	1.45	2
frozen 7-feature law	1.000	0.197	0.220	0.197	1.52	7
linear	1.000	0.206	0.215	0.206	1.96	2
min_sat	1.000	0.206	0.215	0.206	1.96	3

Preregistered outcome = A (cross-domain transfer works). All forms rank the domain-B D-effect perfectly (rho = 1.000, zero monotonicity violations). The 2-param log is best calibrated (MAE/ECE 0.137, slope closest to ideal 1.0) and beats the 7-param frozen law by 0.060 MAE - consistent with Phase 8.

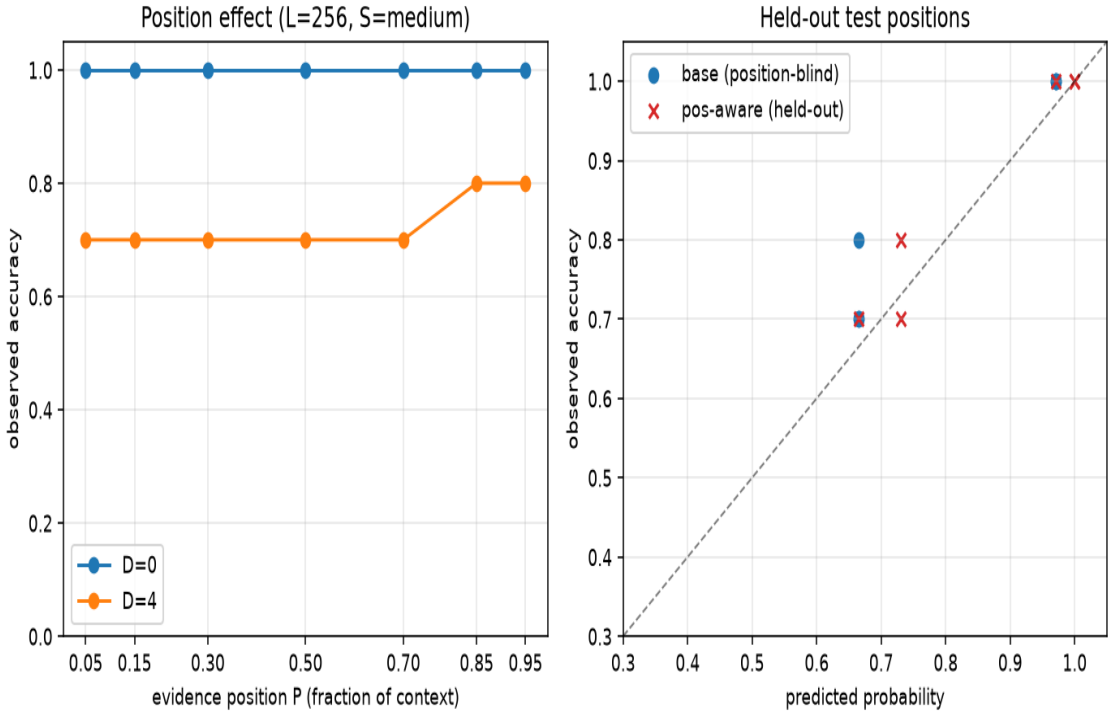
31.2 Target 2 - cross-length (train L=256 -> test L=1024)

S slice	form	Spearman	MAE	ECE	slope
medium	log	1.000	0.163	0.147	1.12
medium	frozen law	1.000	0.169	0.138	0.79
high	log	1.000	0.243	0.227	1.89
high	frozen law	1.000	0.0745	0.0735	0.99
low	log	n/a (const)	0.243	0.243	0.00
low	frozen law	n/a (const)	0.048	0.048	0.00

Honest label. Cross-domain transfer of the retrieval count structure is a clean verdict A: the 2-param log(1+D) form survives a domain change it never saw, with better calibration than the 7-feature law. Cross-length transfer is nuanced: rank structure is domain- and length-robust (rho = 1.000 everywhere non-constant), but calibration depends on which axis is held fixed - the richer law helps where its S/L terms are active (cross-length), the simple log helps where they are not (cross-domain). This reinforces the Phase 9 conclusion R = F(task, f(D), S, L) with condition-dependent coefficients, and adds domain as a condition that changes coefficients, not functional form. Verdicts are preliminary (n=10/cell, 3 D points); higher-n replication is future work.

32. Phase 11 - position effects (zero-refit)

Does the evidence POSITION P (fraction of context before the needle) explain residual error after D, S and L are already known? The protocol was frozen before any cell ran (sha256 1bd50644...): P in {0.05, 0.15, 0.30, 0.50, 0.70, 0.85, 0.95}, D in {0, 4}, L=256, S=medium, n=10 per cell (140 trials), filler held fixed. Isolation is exact: EVERY cell is 267 context tokens, so length and filler amount are constant and only position varies. The position coefficient is fit only on {0.05, 0.30, 0.70, 0.95} and evaluated ZERO-REFIT on the held-out positions {0.15, 0.50, 0.85}.



Phase 11: observed accuracy vs position for D=0 (flat) and D=4 (mild rise at late positions); held-out test observed-vs-predicted, position-blind base vs position-aware extension.

model	MAE (test)	ECE (test)	Spearman	params
base log(1+D), position-blind	0.049	0.049	0.949	2
+ position (pos_edge)	0.027	0.017	0.906	3
frozen 7-feature law	0.060	0.054	0.949	7

Preregistered outcome = B. Adding position improves HELD-OUT calibration (MAE 0.049 -> 0.027, ECE 0.049 -> 0.017; fit-split R2 0.841 -> 0.968, +0.128) but barely moves RANK (Spearman 0.949 -> 0.906), because D=0 is flat (1.00 at every position) and D=4 moves only 0.70 -> 0.80. The MAE gain is 0.021, BELOW the preregistered A threshold of 0.030, so this is NOT an A - no threshold shopping. Position matters primarily as a CALIBRATION feature rather than a rank-ordering feature in this regime, so +P is NOT promoted to the core law; it stays a candidate calibration feature. Core law remains R = F(task, domain, log(1+D), S, L, ...). NOTE: an earlier exploratory exp1_position run at L=512 showed a stronger mid-context effect, but it is non-preregistered and reported only as contextual evidence; it does not contribute to this verdict. Verdicts preliminary (n=10/cell, 7 positions, 2 D).

33. Phase 12A - long-context scaling (zero-refit screening)

Before Phase 12A, long-context scaling - how much of a context window is actually usable - was one of the biggest holes in the evidence (Phases 8-11 all ran at L <= 1024). The protocol was frozen before any cell ran (sha256 d30146e3...): L in {2048, 4096, 8192} x D in {0, 4, 8}, S=medium, position=0.5, n=5 per cell = 45 samples, wiki filler fixed, length token-exact so L is the ONLY variable across the length axis. Short-L baselines reuse existing exp6 (n=20, L in {256,1024}) and exp4 - zero new baseline runs. The run completed 45/45 in 4582s (~76 min), zero failures, batched and resumable by length.

L	D=0	D=4	D=8
2048	1.00	0.80	0.40
4096	1.00	0.40	0.40
8192	1.00	0.00	0.40

Phase 12A screening accuracy (n=5/cell). D=0 stays 1.00 at every length - length alone is harmless; damage appears only under competition.

statistic	value	meaning
harm(2048)	-0.025	essentially no penalty yet
harm(4096)	+0.175	penalty emerging
harm(8192)	+0.375	penalty growing
deltaInt	-0.400	penalty worsens from L=2048 to 8192

Aggregates over D in {4,8} vs the exp6 S=medium L=256 baseline (0.575): the degradation is monotonic in length and conditional on competition.

model	params	log-loss	Brier	Spearman
M_A F(log(1+D), S)	2	0.486	0.0475	0.757
M_B + L	3	0.449	0.0264	0.851
M_C + log(1+D) x L	4	0.4466	0.0260	0.851

M_A -> M_B (adding L) is the major predictive jump (log-loss 0.486 -> 0.449, Brier roughly halved); M_C (adding L x D) adds only a little - most signal is a main effect of L under competition. Frozen rule applied with NO threshold shopping: STAGE A = INTERESTING (harm4096/harm8192 >= 0.15 and deltaInt -0.400 <= -0.15) -> triggers Phase 12B; STAGE B = CONFIRMED AT THE PREREGISTERED SCREENING THRESHOLD (harm8192 0.375 >= 0.20 and deltaInt <= -0.05; n=5/cell is screening-level, not independent confirmation). CONCLUSION: Phase 12A provides strong screening evidence for a competition-dependent long-context penalty and triggers the preregistered Phase 12B confirmation. Long-context degradation is conditional on competing evidence, not length alone - context composition, not raw length, governs reliability. Verdicts preliminary pending 12B (n=10-20, identical frozen spec); S=medium only, S robustness deferred to Phase 13.

34. Phase 12B - long-context confirmation (independent n=20)

Phase 12A gave strong screening evidence and triggered this preregistered confirmation. Phase 12B repeats the IDENTICAL L x D experiment on fresh, independent samples at n=20 per cell, applying the UNCHANGED frozen decision rule. Protocol frozen before any cell ran (sha256 7336f02c..., byte-verified to carry the same thresholds and model spec as 12A): L in {2048, 4096, 8192} x D in {0, 4, 8}, S=medium, position=0.5, n=20/cell = 180 samples, true sample index t in [5,25) - disjoint from 12A's [0,5) (min seed 502053), so no seed collides and the confirmation set is genuinely independent. Run completed 180/180 in 11238s (~187 min), verified, batched and resumable (resumed once after an interrupt with no recomputation or seed drift).

L	D=0	D=4	D=8
2048	1.00	0.55	0.35
4096	1.00	0.60	0.70
8192	1.00	0.40	0.20

Phase 12B confirmation accuracy (n=20/cell, independent samples). D=0 stays 1.00 at every length - length alone is harmless; damage appears only under competition, deepest at L=8192/D=8 (0.20).

statistic	value	meaning
harm(2048)	+0.125	mild penalty
harm(4096)	-0.075	mid-length noise (not in the rule)
harm(8192)	+0.275	penalty at max length
deltaInt	-0.150	penalty worse from L=2048 to 8192

Aggregates over D in {4,8} vs the exp6 S=medium L=256 baseline (0.575). The two statistics that drive the frozen rule both clear: harm(8192) = +0.275 >= 0.20 and deltaInt = -0.150 <= -0.05.

model	params	log-loss	Brier	Spearman
M_A F(log(1+D), S)	2	0.4774	0.02253	0.804
M_B + L	3	0.4723	0.02014	0.797
M_C + log(1+D) x L	4	0.4720	0.02012	0.797

M_A -> M_B again improves prediction (log-loss 0.4774 -> 0.4723); M_C (adding L x D) adds essentially nothing (0.4720) - most of the length signal is a main effect under competition, as in 12A. Frozen rule applied with UNCHANGED thresholds, no threshold shopping: STAGE A = INTERESTING (harm8192 0.275 >= 0.15 and deltaInt -0.150 <= -0.15); STAGE B = CONFIRMED AT THE PREREGISTERED THRESHOLD on fresh n=20 data (harm8192 0.275 >= 0.20 AND deltaInt -0.150 <= -0.05, both True). CONCLUSION: a competition-dependent long-context penalty is CONFIRMED on independent samples - D=0 is 1.00 at every length while the penalty emerges and deepens only under competition, so context COMPOSITION, not raw length, governs reliability. With 12A (n=5 screening) and 12B (n=20 confirmation) both clearing the same frozen rule, the long-context arm rests on a preregistered independent confirmation, not a single screening pass. Caveats: single-cell noise at n=20 (SE ~0.11) and harm4096 = -0.075 is a mid-length non-monotonicity (not in the rule) - the confirmed claim is the aggregate 8192-vs-2048 contrast, not strict per-length monotonicity;

S=medium, one model, cross-model/S robustness deferred to Phase 13.

End of report.